

Question	Answer	Marks
2(a)	rate of change of velocity	B1
2(b)	$\frac{1}{2} m(\Delta)v^2 = mg(\Delta)h$	C1
	$v^2 = 5.9^2 + 2 \times 9.81 \times 7.8$ $v^2 = 188$	C1
	$v = 14 \text{ m s}^{-1}$	A1
	or by resolving components Vertically: $v^2 = u^2 + 2as$ $v^2 = (5.9\sin 60)^2 + 2 \times -9.81 \times (1.2 - 9.0)$ $v_v = 13.4$	(C1)
	horizontally: $v_h = 5.9\cos 60$ $v_h = 2.95$	(C1)
	resultant velocity $= \sqrt{(13.4^2 + 2.95^2)}$ $= 14 \text{ m s}^{-1}$	(A1)
2(c)(i)	(As the diver moves down their) speed decreases	B1
	(So) viscous force / drag (force) decreases	B1
2(c)(ii)	$(F =) \rho g V$ $= 1000 \times 9.81 \times 7.5 \times 10^{-2} = 740 \text{ (N)}$	A1

Question	Answer	Marks
2c(iii)	resultant force = $740 + 950 - (78 \times 9.81)$ = 925	C1
	acceleration = F / m	C1
	= $925 / 78$ = 12 m s^{-2}	A1
	(vertically) upwards	B1

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